# Dosimetry, half-life and statistical analysis of radioactive decays

Jonas Beck and Jonathan Bodky

We measured the local equivalent dose rates in the laboratory and at different distances to a Cs-137 sample. The voltage dependent count rate of a Geiger counter was found to increase rapidly around a certain threshold voltage, above which it became nearly constant. The responsivity to $\gamma$-radiaton of the Geiger counter was measured to be about $0,5\%$. The half life of Ba-137m was found to be $T_{\frac{1}{2}} = (152,9 \pm 1,2)\,\mathrm{s}$. We performed $\chi^2$-tests to analyze the distribution of decay events in an uranium trioxide coated tile for gate times $t_{\mathrm{gate}} \in \{1\,\mathrm{s}; 10\,\mathrm{s}\}$. The assumption that these follow a Poisson distribution could not be rejected at a 5%-significance level. The assumption that the Poisson distribution for $t_{\mathrm{gate}} = 10\,\mathrm{s}$ can be approximated by a normal distribution was rejected at the same significance level.

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## 1 Introduction

Radioactivity describes the spontaneous decay of an unstable nucleus under emission of particles or photons. Despite the health risks of the generated radiation, the application of radioactivity is indispensable in industry and medicine, where strongly located radiation is used to destroy cancerous cells. To avoid unnecessary exposure, it is important to find ways of detecting radiation and quantify its health effects on the human body which requires a good understanding of the decay process.
In this paper, we want to gain a better understanding of radioactive decays, the characteristics of a Geiger counter and the statistics of decay processes.

## 2 Theoretical description of radioactive decay

Radioactive decay is a stochastic process, as it is not possible to determine whether one particular atom will decay in a certain moment or not. It is however possible to calculate the probability for a decay event to occur during a period of time, which allows us to accurately describe the decay behavior of a system with a large number of identical atoms [2].
If we have an initial number $N_0$ of undecayed atoms, the number $N(t)$ of the atoms still remaining at time t is expected to be

$$N(t) = N_0 \cdot e^{-\lambda t}, \tag{1}$$

where $\lambda$ is the decay constant [2]. It is connected to the half life $T_{\frac{1}{2}}$, after which only half of the initially undecayed atoms are left, via

$$\lambda = \frac{ln(2)}{T_{\frac{1}{2}}}. \tag{2}$$

The activity $A(t) = \dot{N}(t)$ follows the same exponential decay law and describes the number of decays per second [2].

## 3 Dosimetry and radiation exposure

As any exposure to ionizing radiation can cause damage to the human body depending on the amount of absorbed radiation, we need a measure to quantify this amount. The most universal measure is the absorbed dose D, defined as

$$D = \frac{dE_{\mathrm{D}}}{dm}, \tag{3}$$

where $dE_{\mathrm{D}}$ is the energy transferred by the radiation to the absorbing material in a volume element with

mass $dm$ [3]. As different kinds of radiation have different biological effects, the more important quantity in radioprotection is the equivalent dose $H$, defined as

$$H = q \cdot D \tag{4}$$

with a weight factor $q$ that depends on the kind of radiation. It ranges from $q = 1$ for $\gamma$- and $\beta$-radiation to $q = 20$ for low energy $\alpha$-radiation. The values for q are chosen such that the equivalent dose $H$ represents the probability of the occurence of stochastic health effects such as cancer [3].
To determine the radiation exposure in the laboratory, we used a Radex RD1503 dosimeter to measure the equivalent dose rate $\dot{H}$ of the background radiation with a weight factor of $q = 1$ as we only expect gamma and beta radiation. The total equivalent dose for a person staying in the laboratory for a whole year can be estimated by $H = t \cdot \dot{H}$ with $t = 1\,a$, assuming that the background radiation stays constant. The obtained value is $H_{\mathrm{lab,1\,a}} = (0,937 \pm 0,040)\,\mathrm{mSv}$. The background radiation exposure in the laboratory does not impose a high danger as the result is just 19 % of the allowed annual limit of $H_{\mathrm{max,1\,a}} = 5\,\mathrm{mSv}$ [3].
For comparison the equivalent dose rate of a Cs-137 sample was measured. The sample was inside an aluminium rod, stored in a lead cube with an iron plug, so after removing the plug a collimated $\gamma$-ray could be used for measurements. The $\beta$-radiation that is also emitted by Cs-137 is blocked by the aluminium. The local equivalent dose rate of the $\gamma$-ray was measured both directly above the lead cube and in a distance of $d = (73,3 \pm 0,5)\,\mathrm{cm}$ and adjusted for the already obtained background radiation as we only want to consider the radiation emitted by the sample. The values obtained are given in Tab. 1 together with the calculated local equivalent dose for $t = 4\,\mathrm{h}$.

| distance | $\dot{H}_{\mathrm{local}}$ $[\frac{\mu\mathrm{Sv}}{\mathrm{h}}]$ | $H_{\mathrm{local,4h}}$ [µSv] |
|---|---|---|
| $\approx 5\,\mathrm{cm}$ | $6,49 \pm 0,12$ | $25,97 \pm 0,50$ |
| $(73,3 \pm 0,5)\,\mathrm{cm}$ | $0,173 \pm 0,045$ | $0,69 \pm 0,18$ |

**Tab. 1:** Local equivalent dose rates and local equivalent doses for different distances to a Cs-137 sample.

The data shows that in a distance of about 1 m to the $\gamma$-source, it would take $t = (2,89 \pm 0,46) \cdot 10^4\mathrm{h}$ to reach $H_{\mathrm{max,1\,a}}$.

# 4 Characteristics of a Geiger-Müller counter

A Geiger-Müller counter is a device used to detect ionizing radiation. The detecting component is the Geiger tube, depicted in Fig. 1. Incoming radiation ionizes the gas in the tube, resulting in free electrons. Those are accelerated by the applied voltage $U_{\mathrm{tube}}$. In the case of low voltages the small number of these primary electrons can not be detected, so we do not get a signal. At a certain threshold voltage $U_{\mathrm{threshold}}$ some of the electrons can cause an electron avalanche near the counting wire due to the high electric field in this region. This leads to a high number of electrons reaching the counting wire, causing a measurable current and the counter is incremented by 1. For even higher voltages, nearly all primary electrons are able to cause an electron avalanche, so the measured counts per time are independent of the exact applied voltage, which is the desired case for the applications of the counter tube [1].

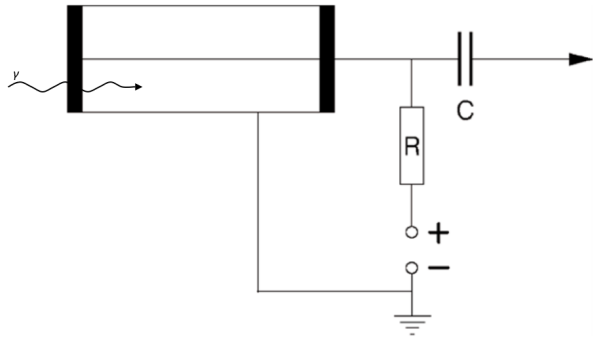


**Fig. 1:** Schematic structure of a Geiger counter. Incoming radiation ionizes the gas in the counter tube on the left which causes an electron avalanche. The emerging power surge is detected by the outer RC-circuit [1].

We examine the characteristics of a Leybold digital counter. For this, we place the opening of the Cs-137 sample in front of the counter tube and measure the detected counts during a gate time of $t_{\mathrm{gate}} = 60\,\mathrm{s}$ for different tube voltages. The results are shown in Fig. 2 and follow the discussed theory. The threshold voltage is $U_{\mathrm{threshold}} = (516 \pm 4)\,\mathrm{V}$. For higher voltages, the count rate seems to increase slowly, which contradicts the theory. We suspect that this slope is in fact caused by statistic scattering around the mean value, which should not change in this region. To further examine this behavior, it would be necessary to perform a series of measurements for every voltage.

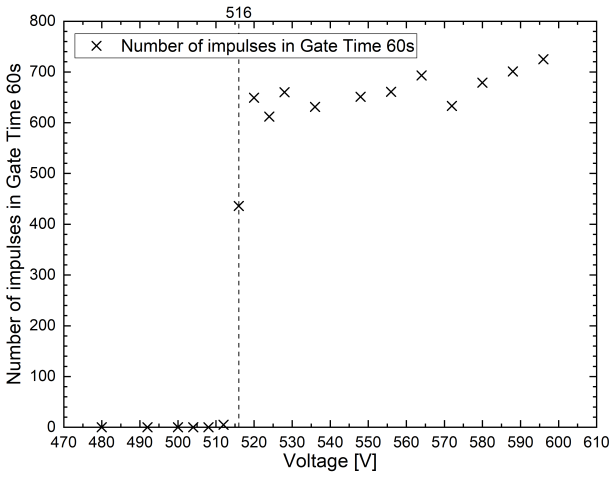


**Fig. 2:** Characteristic curve of a Geiger-counter. For voltages below the threshold voltage of $U_{\text{threshold}} = (516 \pm 4)\,\text{V}$ it does not detect any radiation. Above this voltage the countrate increases rapidly and reaches a plateau, meaning the countrate does not depend on the exact voltage.

## 4.1 Null effect

As even without any particular radioactive materials nearby there is still radiation left due to cosmic radiation and naturally ocurring radioactive isotopes, an unshielded Geiger counter will always detect some events. This has to be taken into account when we are interested in the activity of a particular sample, as we have to adjust the data for the underground. We measure this null effect for the Leybold counter both on the workbench in the laboratory and in a lead cube by counting the detected amount of decays during a time of $t_{\text{gate}} = 5\,\text{min}$ at a tube voltage of $U_{\text{tube}} = 556\,\text{V}$. The resulting values are given in Tab. 2.

| position of the counter | null effect [$\text{min}^{-1}$] |
|---|---|
| standard workbench | $33,2$ |
| inside a lead cube | $10,2$ |

**Tab. 2:** Null effect for a Geiger counter at different positions in the laboratory.

As expected, the null effect inside the lead cube is smaller, being just $30,7\,\%$ that of the null effect measured on a standard workplace due to the shielding effect of the lead.

## 4.2 Responsivity of the Geiger counter

Since a counter tube has finite extent, not every incoming $\gamma$-quantum interacts with the filling gas inside the tube. The ratio between the number of incoming and detected $\gamma$-quanta is the responsivity $R_\gamma$ of the counter to $\gamma$-radiation. We determine this for the Leybold counter by taking the Cs-137 sample out of its lead cube, placing it in a distance of $d = (38,8 \pm 0,5)\,\text{cm}$ to the counter tube and measuring the number of detected quanta during a gate time of $t_{\text{gate}} = 10\,\text{s}$. Assuming the emission of radiation from the sample to be isotropic, we can calculate the responsivity as

$$R_\gamma = \frac{N_{\text{det}}}{A_{\text{Cs}} \cdot \frac{a_{\text{det}}}{4\pi d^2}} \tag{5}$$

with the detected number of decays per second $N_{\text{det}}$, the activity of the sample $A_{\text{Cs}}$ and the sensitive area of the tube $a_{\text{det}}$. As the initial activity of the Cs-137 sample, its half life and its date of acquisition are known, we can calculate the momentary activity as $A_{\text{Cs}} = (1,36372 \pm 0,00059)\,\text{MBq}$. The responsivity is therefore $R_\gamma = (0,670 \pm 0,020)\,\%$.
The obtained value shows that it is very unlikely for a $\gamma$-quant to interact with the gas inside the counter tube.

# 5 Measuring the half-life of Ba-137 m

The used Cs-137 can $\beta$-decay both into the ground state and the excited state of Ba-137 as shown in Fig. 3. The latter then decays into the ground state emitting $\gamma$-quanta. To measure the half life of Ba-137m, we separated it from the Cs-137 sample which is possible due to the different solubility of the two elements. This sample was then put in front of the Leybold Geiger-counter to measure the number of detected decays within a gate time of $t_{\text{gate}} = 10\,\text{s}$.

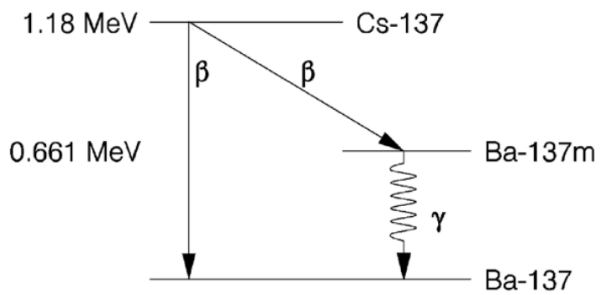


**Fig. 3:** Decay scheme for Cs-137. The isotope decays either into an excited or into the ground state of Ba-137 [1].

The resulting data is shown in Fig. 4. To determine the half life of Ba-137m, we fitted the theoretical expectation

$$A(t) = A_{\text{Null}} + A_0 \cdot e^{-\frac{t}{t_0}} \tag{6}$$

to the data where $A_{\text{Null}}$ accounts for the null effect. The fit parameter $t_0$ allows us to calculate the half life according to eq. 2 with $\lambda = \frac{1}{t_0}$. We receive $T_{\frac{1}{2}} = (152,9 \pm 1,2)\,\text{s}$. This value coincides with the literature value of $T_{\frac{1}{2},\text{theo}} = 153,12\,\text{s}$ [4] within its margin of error.

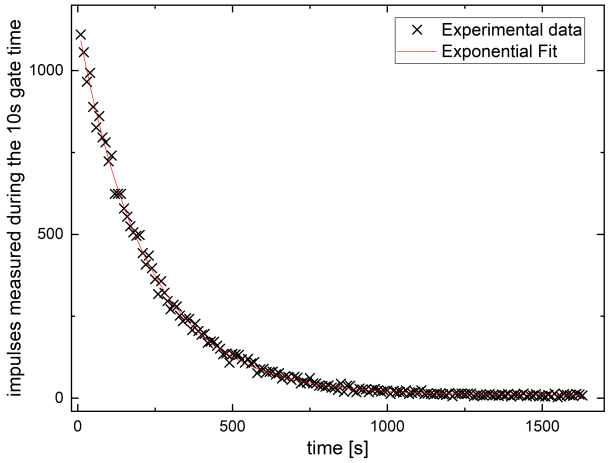


**Fig. 4:** Decay rate of a Ba-137m sample with an exponential fit. The exponent of the fit allows us to calculate the half life of Ba-137m as $T_{\frac{1}{2}} = (152,9 \pm 1,2)\,\text{s}$.

# 6 Statistical analysis of radioactive decays

While the decay law in eq. 1 gives us the expected value for the remaining number of undecayed atoms, the observed number will scatter around this value due to the stochastic nature of radioactive decay. A closer look at the underlying statistics is necessary to analyze data describing radioactive processes.

## 6.1 Poisson and normal distribution

In a given time $t_{\text{Gate}}$ only an integer number of decays can be observed. The probability follows a binomial distribution with the number of atoms $N_{\text{atom}}$ and the probability for one atom to decay during the observation time $p_{\text{decay}}$ as parameters. For the limit of $N_{\text{atom}} \to \infty$, $p_{\text{decay}} \to 0$ with constant mean $\mu = N_{\text{atom}} \cdot p_{\text{decay}}$, the binomial distribution can be approximated by the Poisson distribution. Here the probability of $k$ decays is given as

$$P(k) = \frac{\mu^k}{k!} e^{-\mu}. \tag{7}$$

For a sufficiently large mean $\mu$ the poisson distribution can be approximated by a normal distribution with the standard deviation $\sigma = \sqrt{\mu}$

$$P(k) = \frac{1}{\sqrt{2\pi\mu}} e^{-\frac{(k-\mu)^2}{2\mu}}. \tag{8}$$

## 6.2 Hypothesis tests for Poisson and normal distribution

We examine the quality of these approximations using the Leybold counter and a tile coated with uranium trioxide. We perform a series of measurements for gate times $t_{\text{gate}} = 1\,\text{s}$ and $t_{\text{gate}} = 10\,\text{s}$. The theoretical expectation of a poisson distribution, calculated with the mean $\mu$, is shown together with the experimental data in Fig. 5 for $t_{\text{gate}} = 1\,\text{s}$. As the measurement with a larger gate time leads to a higher mean value, we additionally fit a normal distribution with $\sigma = \sqrt{\mu}$ to the data. The result is shown in Fig. 6.

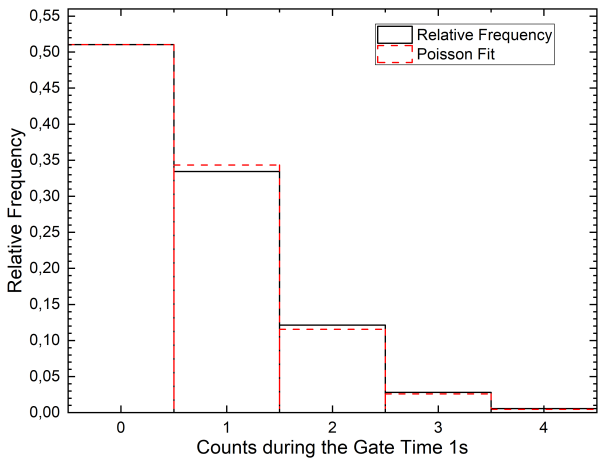


**Fig. 5:** Distribution of the number of counts detected in a $t_{\text{gate}} = 1\,\text{s}$ interval. A Poisson distribution was fitted using the mean value of the experimental data.

To quantify the quality of the approximation as a poisson distribution, we perform a $\chi^2$-test. The data was separated into classes, considering that the theoretically expected value for each class is at least 5. We test the null hypothesis, stating that the data follows a Poisson distribution, at a significance level of 5%. The results are shown in Tab. 3. As the obtained values for $\chi^2$ both lie below their respective threshold, the null hypothesis that the data follows

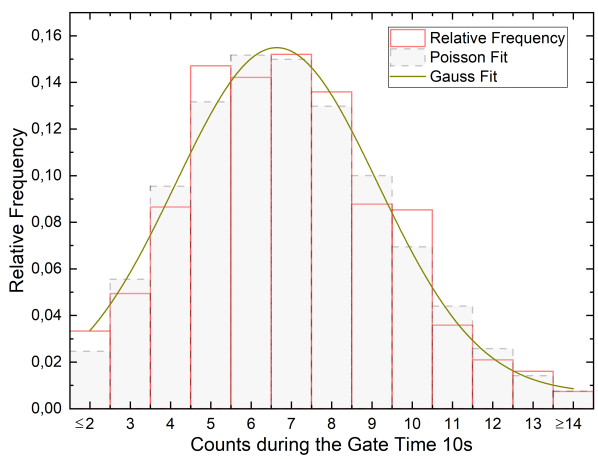


**Fig. 6:** Distribution of the number of counts detected in a $t_{\text{gate}} = 10\,\text{s}$ interval. A Poisson and a normal distribution were fitted using the mean value of the experimental data. For the Gauss fit the standard deviation was set to be the expected value $\sigma = \sqrt{\mu}$, assuming the data follows a Poisson distribution.

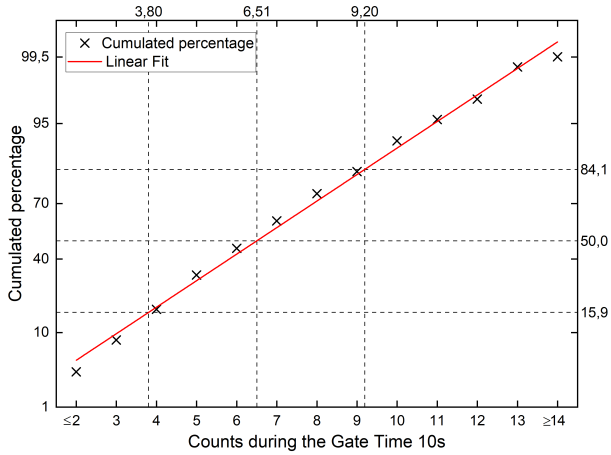


**Fig. 7:** Probability plot of the cumulated relative frequency of measured counts for $t_{\text{gate}} = 10\,\text{s}$ with a linear fit. From this we can read off the mean value and the standard deviation of the Gauss distribution which best describes the experimental data.

a Poisson distribution must be accepted in both cases.

| $t_{\text{gate}}$ [s] | Null hypothesis | $\chi^2$ | $\chi^2_{5\%,\text{max}}$ |
|---|---|---|---|
| 1 | Poisson distribution | $1,10$ | $7,81$ |
| 10 | Poisson distribution | $13,9$ | $21,03$ |
| 10 | normal distribution | $22,8$ | $18,3$ |

**Tab. 3:** Calculated values for $\chi^2$ and threshold values for a significance level of 5%.

The data from the $t_{\text{gate}} = 10\,\text{s}$ mesurement is shown in a probability plot in Fig. 7 together with a linear fit. The fit allows us to read off $\mu = 6,51$ and $\sigma = 2,70$, which deviates by $5,8\,\%$ from the expected value of $\sqrt{\mu}$. For $t_{\text{gate}} = 10\,\text{s}$ we also test the null hypothesis stating the data follows a normal distribution at the same significance level as before. The calculated values are given in Tab. 3. As the obtained $\chi^2$ lies above the threshold value, the null hypothesis that the data follows a normal distribution must be rejected at a 5%-significance level.

## 7 Conclusion

In this experiment, we first measured the equivalent dose rate for the background radiation and for different distances to a Cs-137 sample. We found that the exposure to the background radiation in the laboratory does not impose a high risk, as a person staying there for one year does not exceed the annual limit of $H_{\text{max},1\,\text{a}} = 5\,\text{mSv}$.
We examined the characteristics of of a Geiger counter and found that it only operates above a threshold voltage of $U_{\text{threshold}} = (516 \pm 4)\,\text{V}$. The count rate of the Geiger counter in the absence of particular radioactive material was found to be $33,2\,\text{min}^{-1}$, which could be reduced to $10,2\,\text{min}^{-1}$ by placing the counter in a lead box. The responsivity of the Geiger counter to $\gamma$-quanta was measured to be $R_\gamma = (0,670 \pm 0,020)\,\%$ using a Cs-137 sample with known activity. The half life of Ba-137m was found to be $T_{\frac{1}{2}} = (152,9 \pm 1,2)\,\text{s}$, coinciding with the literature value within its margin of error.
We statistically analyzed the distribution of radioactive decays by performing $\chi^2$ tests for the assumption that the data follows a Poisson distribution. This assumption holds at a significance level of 5% for $t_{\text{gate}} \in \{1\,\text{s}; 10\,\text{s}\}$. The assumption that this Poisson distribution can be approximated by a normal distribution for $t_{\text{gate}} = 10\,\text{s}$ was rejected at a significance level of 5%.

## References

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